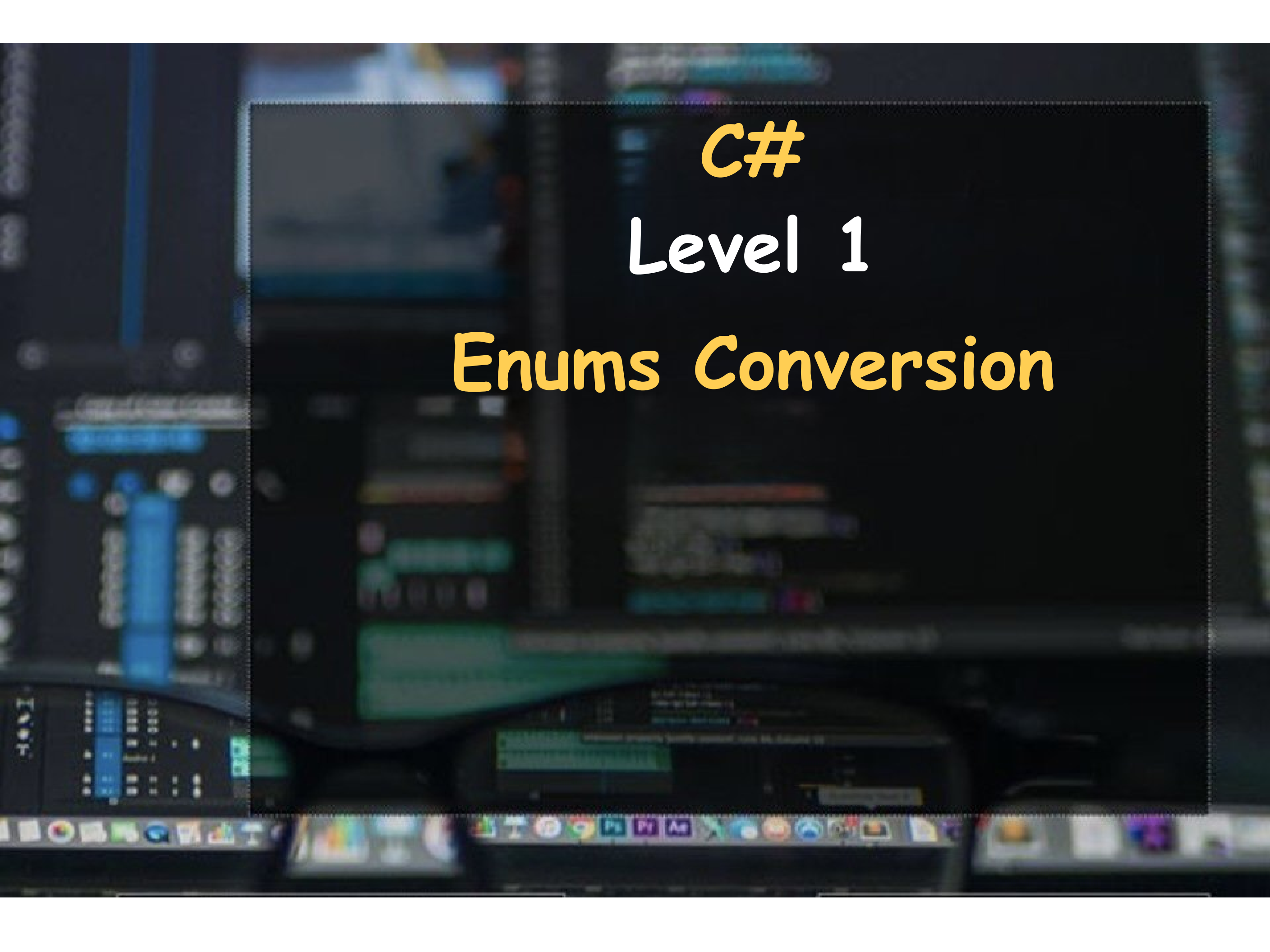




C#

Level 1

Enums Conversion



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Definition

Enum Conversion

-  An ``enum`` has an underlying integral type, usually ``int``.

-  Converting between an ``enum`` and its underlying integer type requires explicit casting.

Explanation

 Enum → int

 Use `(int)` before the enum value.

 By default, enum members start at `0`.

 int → Enum

 Use the enum type in parentheses before the integer.

 The integer is interpreted as the enum's underlying value.

Procedures

1. Create or use an enum.
2. Convert an enum value to `int` using `(int)`.
3. Convert an integer to the enum using `(EnumType)`.
4. Check which enum member corresponds to the value.

💡 Example

```
enum WeekDays
{
    Monday,
    Tuesday,
    Wednesday,
    Thursday,
    Friday,
    Saturday,
    Sunday
}
```

```
Console.WriteLine("Weekday: {0}", WeekDays.Friday);
Console.WriteLine("Day number: {0}", day);
int day = (int)WeekDays.Friday;
Console.WriteLine("Day number: {0}", day);
Console.WriteLine("New weekday by entering int 5: {0}", wd);
```

➡ Result

Weekday: Friday

Day number: 4

New weekday by entering int 5: Saturday

Note

- ◆ Enum numbering starts at `0` unless values are explicitly assigned.
- ◆ Therefore, `Friday` has the value `4`.
- ◆ `(int)WeekDays.Friday` → `4`
- ◆ `(WeekDays)5` → `Saturday`

⊘ Warning

- ◆ An integer does not have to correspond to a named enum member.
- ◆ `(WeekDays)10` is allowed even if `10` is not defined in the enum.
- ◆ In that case, the variable contains the underlying value `10`, but it does not represent a named `WeekDays` member.

Comparison

 Enum $\rightarrow$ int

✓ `(int)WeekDays.Friday`

✓ Result: `4`

 int $\rightarrow$ Enum

✓ `(WeekDays)5`

✓ Result: `Saturday`

✦ Summary

- ✓ Enums normally use `int` as their underlying type.
- ✓ Enum → `int` requires explicit casting.
- ✓ `int` → Enum also requires explicit casting.
- ✓ Default enum values start from `0`.

Thank you 😊